L2a: slope e of counted decomposition work at m = 3 (partial ladder,

s = 6..11) **Lead.** L2a of `research/notes/ecdlp-general/RESEARCH_PRIME_FIELD_LEADS_20261031.md`; thresholds in `../PLAN.md` (interval of e entirely below 0.5 reopens; interval containing 2 closes; null prediction e ≥ 1). **Status.** partial: the frozen ladder is s = 6..12 and the run stopped during s = 12 (host load; the workflow was stopped). **Decision.** does not reopen; the G arm closes under the frozen rule (its interval contains 2); the S arm is above 2 and is reported as observed.

This report was written by the synthesis step (2026-11-01) from the files the lead left: `results/fit.csv`, `results/per_s_means.csv`, `results/L2a_work_vs_s.svg`, `raw/run.log`, `src/l2a.rs`, `src/stats.rs`, and the per-attempt table `attempts.csv` (in the artifact store, `.artifacts.json`, sha256 `f1520b5e…`). No number is new.

1. Design as run (deviations from the lead text named)

- Harness `src/l2a.rs` (standalone Rust, `rustc -O`, no crates). Prime p ≈ 2^{3s}, window [0, 2^s), B = 2^s; one fresh prime per s, 3 curves per prime, 32 targets per curve (96 attempts per s per arm). Question per attempt: do $x_1 \le x_2$, x_3 in the window, each liftable, exist with $R = \pm P_1 \pm P_2 \pm P_3$?
- Arms: **G** group meet-in-the-middle over pairs ($x_1 \le x_2$), four sums $R \mp P_1 \mp P_2$ by x-only addition with batched Montgomery inversion, test x(sum) < B; **S** algebraic S_4: per pair build the quartic $S_4(x_1, x_2, X, x_R)$ in X, find its F_p roots (gcd with X^p − X, Cantor–Zassenhaus), test root < B and liftable; **T** triple scan over all $x_1 \le x_2 \le x_3$ (s ≤ 8 only, an independent check of G and S).
- Deviation: the lead text named the repository's SAT/SMT decomposers as the algebraic arm. The agent used the S_4 root-finding arm instead (it builds and counts in the same unit); the SAT/SMT encodings were not run.
- Metric: Montgomery field multiplications (every one, inside inversions, root finding and window tests included) plus decisions, per attempt. No wall time is recorded; the clock is read only for the 20-minute stop guard, which never fired.
- Fit: ordinary least squares of $\log_2$(mean work over the 96 attempts) on s, bootstrap over attempts (4000 resamples) for the interval. Since B = 2^s and p ≈ B^3, the slope is the e of note §2 (work per attempt = p^{e/3} = B^e).

2. Cells attempted

s	$\log_2$ p	B	G	S	T
6	18.66	64	96 ok	96 ok	96 ok
7	21.54	128	96 ok	96 ok	96 ok
8	24.86	256	96 ok	96 ok	96 ok
9	27.19	512	96 ok	96 ok	not run (s > 8 by design)
10	30.42	1024	96 ok	96 ok	—
11	33.38	2048	96 ok	96 ok	—
12	36.68	4096	32 ok (curve 0 only)	not reached	—

The s = 12 rows (one curve, G arm only) were removed from `attempts.csv` by the resuming session so that a clean rerun of s = 12 could append both arms; that rerun did not happen. Every cell that ran finished (no stop-guard hits); the ladder is incomplete, not failed.

3. Fitted slopes (`results/fit.csv`; interval = bootstrap 2.5 % to 97.5 %)

arm	class	points (s)	e	interval
G	all	6	1.98	[1.94, 2.01]
G	satisfiable	6	2.03	[1.89, 2.33]
G	unsatisfiable	6	1.97	[1.93, 2.00]
S	all	6	2.21	[2.16, 2.24]
S	satisfiable	6	2.24	[2.09, 2.59]
S	unsatisfiable	6	2.20	[2.15, 2.24]
T	all	3	3.21	[3.00, 3.40]

Per-s means (`results/per_s_means.csv`): G grows from $1.0 \cdot 10^4$ multiplications at s = 6 to $1.04 \cdot 10^7$ at s = 11 at 19.6 to 24.7 multiplications per pair; S from $8.6 \cdot 10^5$ to $2.04 \cdot 10^9$ at 2143 to 4037 multiplications per pair (the per-pair cost of the gcd with X^p − X grows with log p, which is the excess of e over 2). Satisfiable attempts are cheaper in level (early exit) and have the same slope. Graph: `results/L2a_work_vs_s.svg` ($\log_2$ mean work against s, one line per arm, slope in the legend).

4. Decision against the frozen rule

- Reopen needs an interval entirely below e = 0.5. The lowest interval measured is [1.94, 2.01]. Not reopened, on any arm or class.
- Close needs an interval containing e = 2. G, all classes: yes (closes for the MITM encoding, which is the enumeration-over-pairs cost e = 2 the note predicts for MITM at m = 3 under §2's "enumeration gives e = m − 1" for the pair stage). S: the interval [2.16, 2.24] lies above 2, so the literal clause does not fire; it is recorded as observed: the algebraic arm costs more than the group arm by a factor that grows like log p, and nothing in it is below the MITM slope. T confirms e ≈ 3 for a full triple scan.
- Null prediction e ≥ 1: met on every arm.

5. Limits

- s = 12 is missing; the slopes rest on six points (G, S) and three (T).
- The SAT/SMT decomposers the lead text named were not measured; the S arm is an algebraic root-finding encoding, not a solver encoding.
- One prime per s and three curves per prime; curve-to-curve spread is inside the bootstrap over attempts, not estimated separately.
- No PDF renderer was available to the lead agent; `REPORT.pdf` was rendered by the synthesis step with Typst in a scratch virtualenv.

6. Reading

At m = 3 the counted work per decomposition attempt grows like B^2 for the group meet-in-the-middle and a little faster than B^2 for the S_4 root-finding arm. The threshold that would beat rho at m = 3 is e < 0.5. Nothing measured is within a factor of four of it in the exponent; the lane stays closed for these encodings, with the s = 12 point still to be taken.

Rerun 2026-11-01: s = 6..12 complete (appended; the sections above

are unchanged) Plan `../rerun_20261101/RERUN_PLAN.md`, report `../rerun_20261101/REPORT.md`, data `../rerun_20261101/L2a/`. The full ladder s = 6..12 was rerun with the same seeds on a quiet host: every s = 6..11 per-s row reproduces the table above exactly, and all 192 s = 12 attempts (G and S) completed inside the 20-minute guard. Fit over seven points:

arm	class	e	interval
G	all	1.97	[1.95, 2.00]
S	all	2.19	[2.16, 2.22]
T	all (s ≤ 8)	3.21	[2.99, 3.40]

Status: complete. **Decision:** does not reopen; G closes (interval contains 2); S above 2, recorded as observed. The SAT/SMT encodings remain unmeasured.